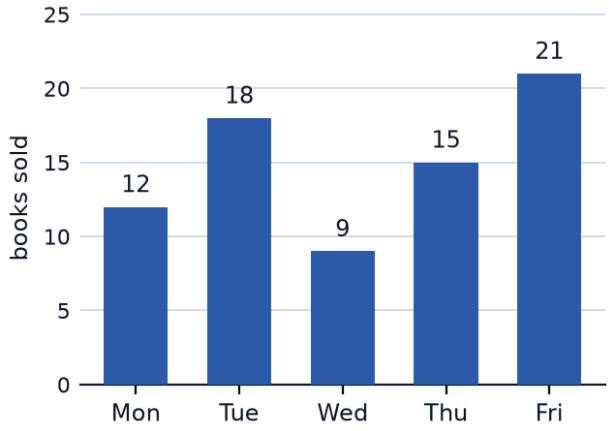
Statistics & Probability

ClipSAT Printable Practice Suite (clipsat.org) • Topic 9.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Reading data from a bar chart

The bar chart shows the number of books sold each day at a bookstore over five days.

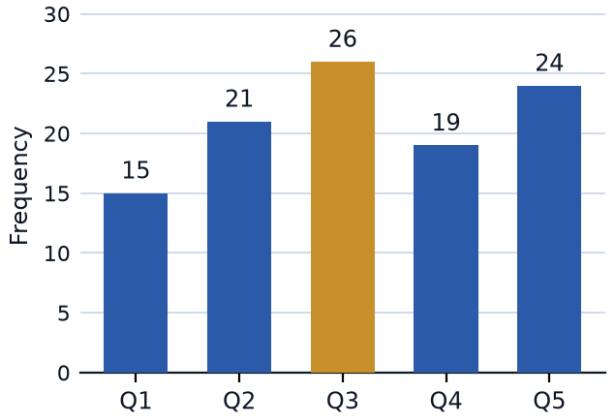


a Find the total number of books sold over the five days.

b Find the mean number of books sold per day.

2 [Foundational] • Target: Mean, median, and mode

Find the mean of $4, 8, 6, 10, 7$.



3 [Foundational] • Target: Mean, median, and mode

Find the median of $3, 9, 5, 1, 7$.

4 [Foundational] • Target: Mean, median, and mode

Find the mode of $2, 4, 4, 6, 8, 4, 2$.

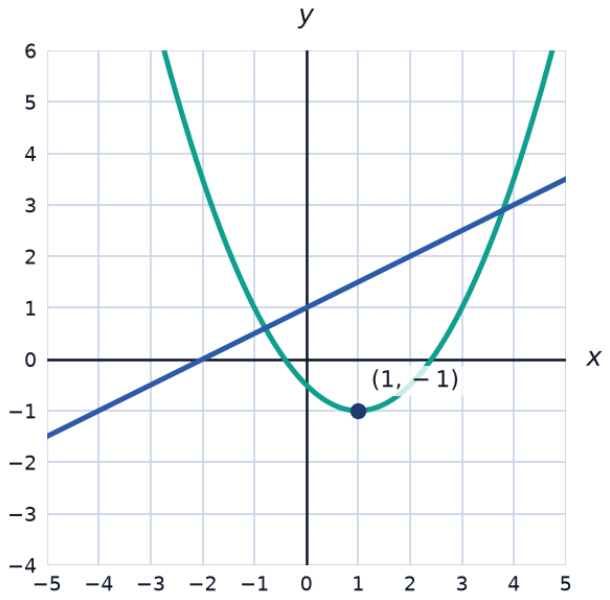
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5 [Foundational] • Target: Mean, median, and mode

A data set has values $8, 12, 10, 15, 10, 14, 11$. Find the mean, median, and mode.

6 [Foundational] • Target: Range and quartiles

Find the range of $5, 12, 8, 20, 3$.



7 [Foundational] • Target: Range and quartiles

For the data set $2, 4, 5, 7, 8, 9, 12$, find Q_1 and Q_3.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Range and quartiles

Find the interquartile range (IQR) for the same data set.

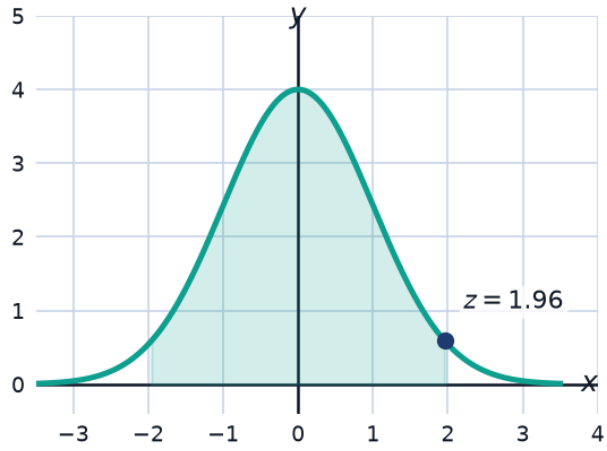
9 [Intermediate] • Target: Variance and standard deviation

Find the variance of $2, 4, 6, 8, 10$, using $\sigma^2 = \frac{\sum(x-\mu)^2}{n}$.

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10 [Intermediate] • Target: Variance and standard deviation

Find the standard deviation of the same data set, rounded to two decimal places.



11 [Intermediate] • Target: Variance and standard deviation

A data set has variance 25. State its standard deviation.

12 [Intermediate] • Target: Frequency tables

A frequency table shows value 1 with frequency 3, value 2 with frequency 5, and value 3 with frequency 2. Find the mean.

13 [Intermediate] • Target: Frequency tables

Using the same frequency table, find the mode.

14 [Intermediate] • Target: Frequency tables

Using the same frequency table, find the median.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Basic probability • Cross-Lesson: 8.01 Exponentials & Logarithms

A fair six-sided die is rolled. Find $P(\text{number} > 4)$.

16 [Challenging] • Target: Basic probability • Cross-Lesson: 1.01 Algebra & Equations

A bag contains 5 red and 3 blue marbles. Find $P(\text{blue})$.

17 [Challenging] • Target: Basic probability • Cross-Lesson: 2.01 Functions & Transformations

A card is drawn from a standard 52-card deck. Find $P(\text{king})$.

18 [Challenging] • Target: Basic probability • Cross-Lesson: 3.01 Sequences & Series

Two fair coins are flipped. Find $P(\text{both heads})$.

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19 [Challenging] • Target: Complementary and mutually exclusive events • Cross-Lesson: 4.01 Trigonometry

If $P(A) = 0.35$, find $P(\text{not } A)$.

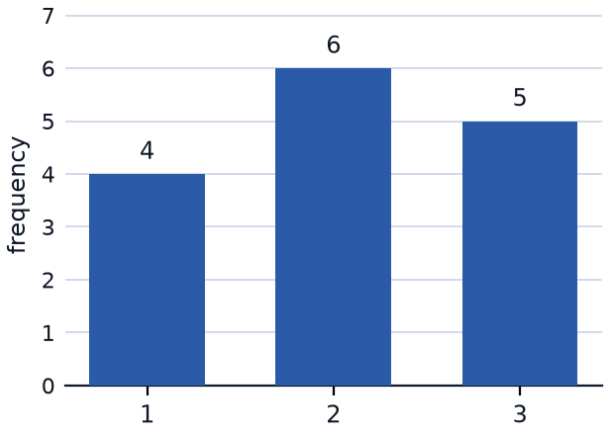
20 [Challenging] • Target: Complementary and mutually exclusive events • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

A die is rolled. Events $A = \{\text{even}\}$ and $B = \{\text{rolling a } 5\}$. Are A and B mutually exclusive?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Frequency tables

The bar chart shows a frequency table. Find the total number of data points.



a. 15

b. 6

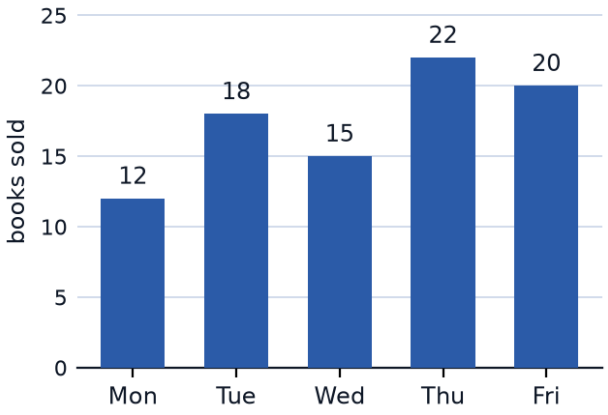
c. 3

d. 10

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22 [Foundational] • Target: MCQ — Reading data from a bar chart

The bar chart shows the number of books sold each day at a bookstore over five days. Find the total books sold.



a. 87

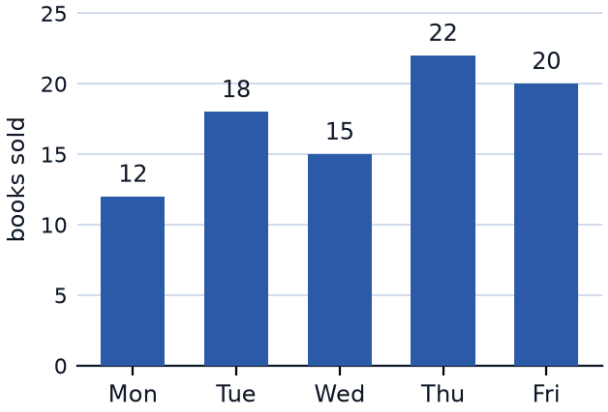
b. 22

c. 77

d. 97

23 [Foundational] • Target: MCQ — Reading data from a bar chart

Using the same bar chart, find the mean number of books sold per day.



a. 15

b. 18

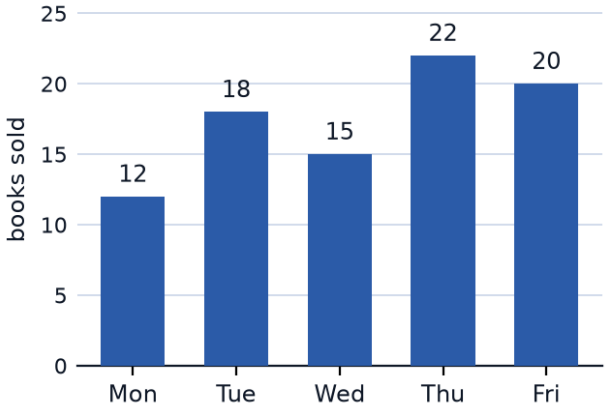
c. 20

d. 17.4

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24 [Foundational] • Target: MCQ — Reading data from a bar chart

Using the same bar chart, find the range of books sold.



a. 12

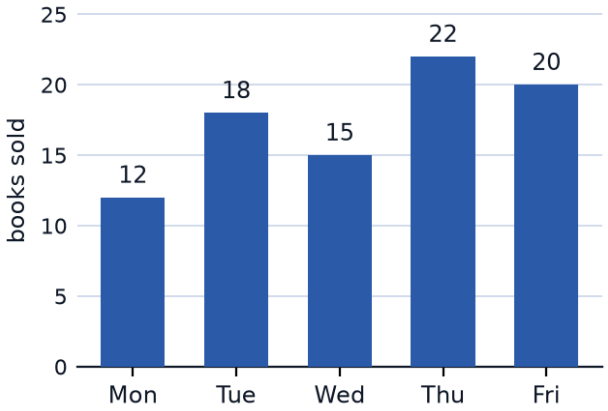
b. 22

c. 8

d. 10

25 [Foundational] • Target: MCQ — Reading data from a bar chart

Using the same bar chart, on which day were the most books sold?



a. Wednesday

b. Thursday

c. Tuesday

d. Friday

26 [Foundational] • Target: MCQ — Mean, median, and mode

Find the mean of $5, 9, 7, 11, 8$.

a. 8

b. 7

c. 40

d. 9

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27 [Foundational] • Target: MCQ — Mean, median, and mode

Find the mean of $12, 18, 15, 21, 14$.

a. 15

b. 16

c. 80

d. 18

28 [Foundational] • Target: MCQ — Variance and standard deviation

A data set has variance 36. State its standard deviation.

a. 6

b. 12

c. 18

d. 1,296

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Basic probability

A fair six-sided die is rolled. Find $P(\text{number} < 3)$.

a. $\frac{1}{6}$

b. $\frac{1}{2}$

c. $\frac{1}{3}$

d. $\frac{2}{3}$

30 [Intermediate] • Target: MCQ — Complementary and mutually exclusive events

If $P(A) = 0.28$, find $P(\text{not } A)$.

a. 0.28

b. 1.28

c. 0.5

d. 0.72

31 [Intermediate] • Target: MCQ — Independent events

A fair coin is flipped twice. Find $P(\text{both heads})$.

a. $\frac{1}{3}$

b. 1

c. $\frac{1}{4}$

d. $\frac{1}{2}$

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32 [Intermediate] • Target: MCQ — Conditional probability

A box has 12 balls: 7 red and 5 blue. Two balls are drawn without replacement. Given the first is blue, find $P(\text{second is blue})$.

a. $\frac{5}{11}$

b. $\frac{4}{11}$

c. $\frac{5}{12}$

d. $\frac{4}{12}$

33 [Intermediate] • Target: MCQ — Counting: permutations and combinations

A code uses 4 distinct digits from 0–9, where order matters. How many possible codes are there?

a. 10,000

b. 40

c. 210

d. 5,040

34 [Intermediate] • Target: MCQ — Word problems: statistics and probability in context

A survey of 250 students found that 150 like math and 100 like science, with 40 liking both. Find the probability that a randomly selected student likes math or science.

a. 0.84

b. 0.6

c. 0.16

d. 1.0

35 [Intermediate] • Target: MCQ — Variance and standard deviation

Find the mean of $2, 6, 1, 13, 8$.

a. 30

b. 6

c. 7

d. 5

36 [Intermediate] • Target: MCQ — Frequency tables

A bag contains 6 red and 3 blue marbles. Find $P(\text{blue})$.

a. $\frac{3}{11}$

b. $\frac{2}{3}$

c. $\frac{4}{9}$

d. $\frac{1}{3}$

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37 [Intermediate] • Target: MCQ — Basic probability

Find the range of $28, 2, 14, 37, 32$.

a. 35

b. 37

c. 38

d. 33

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Complementary and mutually exclusive events • Cross-Lesson: 3.01 Sequences & Series

Find the median of $2, 19, 28, 15, 50$.

a. 19

b. 22

c. 17

d. 21

39 [Challenging] • Target: MCQ — Independent events • Cross-Lesson: 4.01 Trigonometry

Two independent events have $P(A) = 0.7$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.

a. 1.2

b. 0.35

c. 0.3

d. 0.4

40 [Challenging] • Target: MCQ — Conditional probability • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

If $P(A) = 0.2$, find $P(\text{not } A)$.

a. 0.7

b. 0.9

c. 0.8

d. 0.2

41 [Challenging] • Target: MCQ — Counting: permutations and combinations • Cross-Lesson: 6.01 Limits & Derivatives

A data set has variance 100. State its standard deviation.

a. 9

b. 11

c. 100

d. 10

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42 [Challenging] • Target: MCQ — The five-number summary • Cross-Lesson: 7.01 Integration & Applications

A box has 10 balls: 4 red and 6 blue. Two balls are drawn without replacement. Find $P(\text{2nd is red} \mid \text{1st is red})$.

a. $\frac{4}{9}$

b. $\frac{2}{3}$

c. $\frac{1}{3}$

d. $\frac{2}{5}$

43 [Challenging] • Target: MCQ — Reading data from a bar chart • Cross-Lesson: 8.01 Exponentials & Logarithms

How many ways can a committee of 2 be chosen from 6 people (order doesn't matter)?

a. 17

b. 30

c. 13

d. 15

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Algebra & Equations

A bag contains 7 red and 4 blue marbles. Find $P(\text{red})$.

a. $\frac{7}{11}$

b. $\frac{1}{11}$

c. $\frac{4}{11}$

d. $\frac{7}{4}$

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Functions & Transformations

Two independent events have $P(A) = 0.8$ and $P(B) = 0.4$. Find $P(A \text{ and } B)$.

a. 0.37

b. 0.32

c. 1.2

d. 0.27

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